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MASTER'S STUDY PROPOSAL

隱形帳簿：量化印尼平台經濟中的隱形缺口
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

The digital economy is measured through several indicators (platform transaction value, platform revenue, national e-commerce statistics and payment data) that capture different parts of the same activity but are often used interchangeably. This study asks how much economic activity remains invisible when digital platforms are measured through their reported revenue, and when that choice changes the conclusion about growth. Between 2022 and 2023, Tokopedia's revenue rose 53.2 percent while its transaction value fell 8.9 percent. Among Indonesian platforms reporting both figures, revenue grew faster in six of eight year-to-year comparisons, by a margin significantly larger than among listed platforms outside Indonesia. Nationally, most e-commerce growth came from more participating businesses, and 98.5 percent of 2023–2024 growth lay outside the marketplace channel of BPS-Statistics Indonesia. Conclusions about scale and growth therefore depend on the indicator used.

1. Introduction

Between 2022 and 2023, Tokopedia's revenue rose 53.2 percent while the transaction value it processed fell 8.9 percent: read through its accounts, the platform was expanding quickly; read through its transactions, it was shrinking. Both figures come from the same company's disclosures. Of the increase in net revenue, 60.6 percent reconciles to lower customer incentives, the partner incentives and end-user promotions that GoTo deducts from gross revenue to arrive at net revenue (GoTo Gojek Tokopedia 2024), and 39.4 percent to higher gross revenue. The reversal is not isolated: among the three platforms that report both figures themselves, revenue grew faster than transaction value in six of eight year-to-year comparisons between 2020 and 2025, and in two of these the measures moved in opposite directions.

The same gap runs through the wider digital economy. Southeast Asia's digital economy grew from approximately US\$100 billion in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). That figure is one indicator of the activity: the gross value of the transactions involved. Digital platforms, which carry many of those transactions, report a different number, their own revenue; national statistical agencies report a third, the online sales of surveyed businesses; and payment systems report a fourth, the value of funds moved. The indicators are designed to measure different things, yet they are often used interchangeably, and substituting one for another changes the measured scale, growth and even the direction of digital activity.

Suppose a platform processes 100 units of transaction value but recognizes only 10 as its own revenue: the commissions and fees it earns for arranging the transactions. Depending on the business model, the remaining 90 can include merchant receipts, driver payouts, inventory costs, taxes and other pass-through payments. An analyst using the platform's accounts sees 10, while one measuring the commerce it processes sees 100. I call the difference the invisible wedge (W). The term refers to what a revenue-only measure excludes, not to unrecorded activity. Revenue records what the platform earned, while transaction value records what it carried. Platform economics and revenue-recognition rules explain why such a gap exists (Rochet and Tirole 2003; Hagiu and Wright 2015; Evans and Schmalensee 2016; International Accounting Standards Board 2014), but there is little evidence on how large it becomes in practice or whether it stays stable enough for revenue to stand in for the commerce a platform carries.

Against this backdrop, I study Indonesia, the largest digital market in Southeast Asia. Platform reports disclose both the commerce a platform carries and the revenue it recognizes, so the consequences of substituting one for the other can be observed directly within the same accounts. The study covers four marketplaces, including Tokopedia, Blibli, Bukalapak and Shopee, and Grab, whose Indonesian business is mainly delivery and other on-demand services. Tokopedia, Blibli and Bukalapak report both measures themselves (i.e., issuer-reported figures). For Grab and Shopee, Indonesian figures must be constructed from company-wide ratios and a third-party estimate.¹ To limit the concerns this raises, I follow each platform against itself through time and keep issuer-reported and constructed figures apart, so that the company, the business activity and the reporting convention stay as constant as the disclosures allow. I then compare the platform evidence with two independent indicators: national e-commerce statistics from BPS-Statistics Indonesia (BPS), the national statistical agency, and payment statistics from Bank Indonesia, the central bank. Only platform transaction value and revenue enter the wedge; BPS and payment data serve as comparison indicators with their own definitions and scope.

In 2023, Tokopedia, Shopee and Grab together processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, a wedge of US\$40.07 billion. The wedge establishes scale; the central question is whether revenue remains a stable guide to the commerce underneath it,

¹ See Section 4.3 for more details.

and, as Tokopedia shows, it need not. National statistics show the same problem one level up. BPS classifies each surveyed business's online sales by the channel used, such as marketplaces, social media, instant messaging or its own website. It places the 2023 marketplace component of e-commerce at about US\$13.2 billion, below the transaction value of either Tokopedia or Shopee alone, and reports that 98.5 percent of the increase in e-commerce value between 2023 and 2024 came from outside its marketplace category. Payment measures from Bank Indonesia grew between 1.8 and 11 times as fast as e-commerce value.

One concern is that some of the Indonesian figures are constructed rather than reported by the platforms (Grab and Shopee). I address this in three ways. First, the growth analysis uses only the platforms that report both transaction value and revenue themselves. Second, I vary the monetization rate assumed for Shopee. Third, I compare the Indonesian platforms with eight listed platforms outside Indonesia: revenue also grew faster in 22 of 29 comparisons there, but the median divergence was 7.10 percentage points, against 38.92 in Indonesia, a difference that is statistically significant (Section 5.1).

A statement about how large Indonesia's digital economy is, or how fast it grew, is therefore a statement about an indicator. The choice matters to investors, who value platforms largely on reported revenue; to statistical agencies sizing the digital economy; and to tax authorities, which increasingly rely on marketplace records to identify sellers and their transactions (Ministry of Finance of the Republic of Indonesia 2025; Business Times 2026). An investor relying on platform revenue and an analyst relying on marketplace figures can reach opposite conclusions about the same years because the indicators measure different objects.

1.1 Research question and objectives

One question drives the proposal: *How much economic activity remains invisible when digital platforms are measured through their reported revenue, and when does that choice change the conclusion about growth?* Indonesia provides the main empirical setting. Three objectives of this study follow. First, I measure the invisible wedge across the platform histories available from 2020 to 2025. Second, I reconcile selected large movements with disclosed changes in monetization, customer incentives, revenue recognition and reporting scope. Third, I compare the platform evidence with national e-commerce and payment indicators to test whether they give the same account of economic scale and growth.

A separate comparison of eight listed platform businesses outside Indonesia, such as eBay and Mercado Libre, tests whether the separation between transaction value and revenue appears under other business models.

1.2 Contributions

My study contributes to the literature in several ways. First, I extend the literature on digital platforms. Platform economics and revenue-recognition research explain why a gap between transaction value and revenue exists and can move (Rochet and Tirole 2003; Hagiu and Wright 2015; International Accounting Standards Board 2014), but not how large it becomes in practice. I turn the invisible wedge from a one-year estimate into a measure tracked over time. Revenue is not a stable guide to the commerce behind it: among the issuer-reported platforms, revenue outgrew transaction value in six of eight comparisons, and in two of these the measures moved in opposite directions.

Second, I reconcile a major movement in the wedge using accounting components the platform discloses. Of the increase in Tokopedia's net revenue between 2022 and 2023, 60.6 percent reconciles to lower customer incentives and 39.4 percent to higher gross revenue, even though its transaction value fell that year.

Third, I extend the comparison beyond company accounts, in line with official frameworks that distinguish the commerce facilitated through a platform from the platform's own service (Ahmad and Schreyer 2016; OECD 2023). National statistics and payment indicators disagree with the platform picture, and with each other, about both the level and the growth of Indonesian digital commerce.

Finally, I address a measurement problem in finance. Platform reports reveal both the commerce a platform carries and the revenue it recognizes, allowing the consequences of substituting one measure for the other to be observed directly. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter, while Berg et al. (2020) show that intermediary-held digital traces contain information conventional records can miss. Studies of internet firms also find that activity measures carry information beyond reported revenue (Trueman, Wong and Zhang 2001; Rajgopal, Venkatachalam and Kotha 2003). This study applies those ideas to platform economies and shows that the indicator chosen to describe digital activity changes the economic conclusion.

2. Literature Review

Digital platforms sit between economic activity and the indicators used to measure it. They facilitate transactions between buyers and sellers, recognize only part of that activity as their own revenue, and generate records that can also appear in payment, statistical, and administrative systems. Existing research explains each part of this process from a different angle. Taken together, it provides the foundation for asking how much activity remains invisible when platforms are measured through reported revenue.

2.1 Platform Economics and Revenue Recognition

Platform economics explains why the value of commerce processed through a platform can be much larger than the revenue the platform records. Digital platforms often connect consumers with merchants or other service providers and earn commissions, fees, advertising, logistics, or other intermediation revenue rather than the full value of every transaction they facilitate (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016).

In particular, revenue-recognition rules affect how much of this difference appears in the accounts. Under International Financial Reporting Standard 15 (IFRS 15), whether a company acts as a principal or an agent affects whether it recognizes the full customer payment or only the amount it earns from arranging the transaction (International Accounting Standards Board 2014). Customer incentives, monetization, service mix, acquisitions, and changes in business scope can also move reported revenue without an equivalent change in transaction activity. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when financial information is interpreted. Overall, the literature explains why a transaction–revenue gap can exist and move over time, but not how economically large that gap becomes in practice.

2.2 Measuring the Digital Economy

Official measurement frameworks already distinguish between the commerce facilitated through a platform and the service provided by the platform itself. National-accounting guidance separates the buyer-seller transaction from the platform's intermediation service rather than treating the full transaction value as platform output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025).

In practice, the quantities are published by different producers. For instance, industry reports size the regional digital economy by gross merchandise value (Google, Temasek and Bain 2024), national statistics measure the online sales of surveyed businesses (BPS-Statistics Indonesia 2025), and listed platforms report the revenue they recognize under IFRS 15 (International Accounting Standards

Board 2014). A difference that is harmless when the measures move together becomes important when they do not, because the choice of measure can change conclusions about economic scale and growth. Overall, this study treats substitution between indicators as a question to be tested, as research on night-time lights does for measured income (Henderson, Storeygard and Weil 2012).

2.3 Digital Records and Third-Party Information

Transactions that do not enter a platform's revenue can still leave detailed records within the platform's system. Merchants and service providers can be small, self-employed, or weakly represented in conventional financial records, yet the transactions they make on a platform are recorded by it. This differs from shadow-economy settings, where reliable formal records may be sparse (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Platform-mediated activity can also create traces outside conventional records (Berg et al. 2020; Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

Third-party information changes compliance and enforcement (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The Model Rules of the Organisation for Economic Co-operation and Development (OECD 2020) and the European Union's seventh Directive on Administrative Cooperation (DAC7; European Union 2021) apply this principle to platforms, as does Indonesia's Minister of Finance Regulation No. 37 of 2025 (PMK 37/2025) through seller identification, transaction-linked reporting and marketplace withholding (Ministry of Finance of the Republic of Indonesia 2025). Overall, the relevance here is informational: what a measure or rule can show depends on the records it uses.

2.4 Research Gap

Existing research explains the pieces of the problem separately. Platform economics explains why transaction value can exceed revenue; accounting explains why the gap can move; official measurement frameworks explain why these quantities should not be treated as equivalent; and research on digital records explains why some platform-mediated activity outside revenue can remain observable in platform records.

However, there is a research gap regarding how large these differences become in practice, whether they persist over time, and whether choosing one indicator instead of another changes the economic conclusion. This study addresses that gap by measuring the invisible wedge over time, examining disclosed reasons for major movements, comparing platform accounts with national e-commerce and payment indicators, and using external platform evidence to test whether the same pattern appears beyond Indonesia.

3. Theoretical Framework and Hypotheses

Digital platform activity can be observed through several indicators, but those indicators do not measure the same economic object. The framework begins with the difference between the value a platform processes and the revenue it recognizes, then asks whether that relationship remains stable over time and whether other indicators produce the same account of digital activity. Figure 1 illustrates the framework.

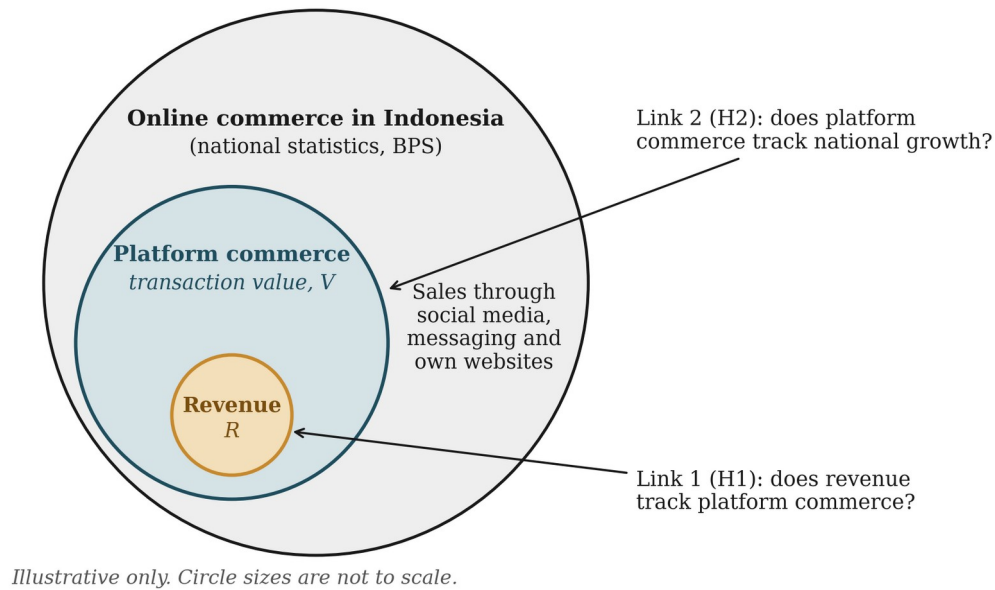


Figure 1. Research framework.

For illustrative purposes only; circle sizes are not to scale, and the platform circle is not identical to the marketplace channel of BPS-Statistics Indonesia, whose definition differs. H1 tests the inner link between revenue and platform commerce; H2 tests the outer link between platform commerce and national e-commerce growth.

3.1 Measurement Framework

For the platform analysis, transaction value and revenue are paired only when they refer to the same period and business activity. Figures that issuers report and figures that must be constructed are kept apart, as are national statistics and payment indicators (Section 4). In my study, I measure the level of the “invisible wedge,” denoted by W , in currency units as follows:

$$W_{it} = V_{it} - R_{it} \quad (1)$$

where V_{it} denotes transaction value processed by platform i in period t , and R_{it} the revenue recognized over the same matched period and business scope. W represents the invisible wedge, calculated as the transaction value occurring outside the platform’s recognized revenue. It should not be interpreted as an accounting omission, as both quantities may be reported within the same company disclosure. It is a transaction-value measure rather than an income or value-added measure; Section 7 states the corresponding inference boundaries. To account for differences in platform size, I also report the wedge relative to revenue. Specifically, I define the *Ecosystem Ratio* (E) as:

$$E_{it} = W_{it} / R_{it} \quad (2)$$

The *Ecosystem Ratio* is constructed for this study as a descriptive transformation of the monetization rate, and its interpretation draws on two established results. First, platform economics shows that booked revenue need not track the transaction value a multisided platform coordinates (Rochet and Tirole 2003; Hagiu and Wright 2015; Evans and Schmalensee 2016). Second, accounting research shows that the share of transaction value recognized as revenue depends on how revenue is recognized, including whether the platform acts as principal or agent (International Accounting Standards Board 2014; De Franco, Kothari and Verdi 2011). For example, an E of 9 means that nine units of transaction value remain outside the revenue for every unit that is recognized as platform revenue. Equivalently, transaction value is ten times revenue, and approximately 10 percent of transaction value is recognized as revenue.

A single ratio does not show whether the relationship holds through time, so I further define annual “growth divergence,” denoted by D , as follows:

$$D_{it} = g(R_{it}) - g(V_{it}) \quad (3)$$

where $g(X)$ is the percentage growth of X from one year to the next. A positive D means revenue grew faster than transaction value, so the platform's monetization rate, $m = R/V$, rose and E fell; a negative D means the reverse. Because $R = m \times V$, any divergence between the two growth rates reflects a change in m (for small changes, D approximates the growth of m), and the same definitions give $E = (V - R)/R = V/R - 1 = 1/m - 1$. If revenue were a stable proxy for transaction value, D would remain close to zero. W gives the wedge in currency units, E expresses it relative to revenue, and D carries the central test: whether the relationship between revenue and transaction value remains stable over time.

3.2 Hypothesis Development

Two hypotheses follow from the research question. H1 asks whether revenue tracks the commerce carried within a platform; H2 asks whether platform records can track the growth of national e-commerce. Together they test the two links between the indicator most often used, platform revenue, and the activity it is used to represent. Table 1 pairs each hypothesis with its principal test.

Platform economics and accounting both suggest that the share of transaction value a platform recognizes as revenue is not fixed. Revenue depends on monetization, customer incentives and whether the platform acts as principal or agent under IFRS 15, and these can change with the platform's business model, scope and pricing (Rochet and Tirole 2003; Hagiu and Wright 2015; International Accounting Standards Board 2014). Research on key performance indicators likewise finds that a disclosed measure becomes less informative when the way it is computed changes over time (Givoly et al. 2019). Revenue therefore tracks transaction activity only while m stays constant, that is, while D is close to zero.

If the relationship between transaction value (V) and platform revenue (R) were stable, revenue would be a workable stand-in for the commerce a platform carries, the growth divergence (D) would remain close to zero and the *Ecosystem Ratio* (E) would be constant. Accordingly, I propose the following hypothesis:

H1. Transaction value and revenue recognized by the same platform grow proportionally over time.

The test compares the size of D with its size among listed platforms outside Indonesia, where revenue and transaction value broadly move together (Section 4.5).

National e-commerce value can grow because existing businesses sell more (the intensive margin) or because more businesses sell online (the extensive margin; Hummels and Klenow 2005). The distinction matters for measurement: growth driven by new sellers can take place in channels that platform accounts do not cover, such as social media and instant messaging, which BPS reports are widely used (BPS-Statistics Indonesia 2025), so platform figures can miss it even when each platform reports accurately. Because BPS reports both the number of e-commerce businesses and total e-commerce value, national growth can be divided arithmetically into these two components (Section 4.4). I therefore hypothesize that:

H2. The growth of national e-commerce value is attributable primarily to an increase in the number of participating businesses, rather than to higher sales per business.

The split is arithmetic and does not assign new businesses to particular channels. H1 and H2 are therefore linked: H1 tests whether revenue represents the commerce a platform carries, and H2 tests whether the commerce platforms carry represents national e-commerce growth.

Two further analyses support these hypotheses without forming separate hypotheses. First, BPS's split of e-commerce value by sales channel shows whether national growth took place inside or

outside the marketplace channel. Second, BPS's business-level comparison shows whether marketplace sellers are more likely to keep financial statements, which bears on the information that platform records can supply to tax administration. Both are reported in Section 5.2 as additional analyses.

Table 1. Hypotheses and principal empirical tests

This table pairs each hypothesis with the principal empirical test used to examine it. H1 concerns transaction value and revenue within platforms, and H2 the national e-commerce evidence. *D* denotes annual growth divergence, defined in Section 3.1. BPS denotes BPS-Statistics Indonesia. None of the tests is interpreted as a causal claim.

Hypothesis	Principal empirical test
H1 - Transaction-revenue divergence	Growth divergence <i>D</i> and the correlation of revenue and transaction-value growth for the platforms that report both figures, benchmarked against 29 comparisons outside Indonesia
H2 - Sources of e-commerce growth	BPS national e-commerce value split into business count and sales per business

4. Data and Methodology

My study uses several indicators because no single one captures every part of digital platform activity. Platform reports provide transaction value and revenue; BPS-Statistics Indonesia provides national e-commerce and business evidence; Bank Indonesia provides payment statistics; and platform companies outside Indonesia provide a separate comparison.

4.1 Selecting the Platform Evidence

The basic observation is one platform in one year, with transaction value and revenue referring to the same period and, as closely as possible, the same business activity. An observation is retained only when its period, geographic and business coverage, units, definitions, and source can be identified; measures referring to different periods are excluded, and major changes in accounting definitions or business structure are recorded.

The main sample consists of the platforms that report transaction value and revenue themselves: Tokopedia, Blibli and Bukalapak. Grab and Shopee, whose Indonesian figures must be constructed, support the 2023 scale comparison.² According to 2023 statistics, Shopee and Tokopedia together accounted for approximately 70 percent of Indonesia's estimated marketplace transaction value (Momentum Works 2024; see Section 4.3). Grab, a major delivery intermediary, was also included in the analysis. As shown in Table 2, the main sample has 11 platform-years and 8 year-to-year comparisons; adding Grab and Shopee raises these to 17 and 12. Each series covers only years for which a matched pair exists.³

² Lazada and TikTok Shop, both large in Indonesia, are outside the sample because no matched Indonesian revenue series has been located for either. The platforms are disclosing cases rather than a representative sample of Indonesian platforms.

³ Tokopedia's 2021 pair covers mismatched periods, and after Tokopedia was combined with TikTok Shop's Indonesian e-commerce business on 31 January 2024, with TikTok taking the controlling stake, the listed parent GoTo reports a contractual fee rather than platform revenue. Blibli's third-party segment includes online travel; its 2020 prospectus pair is used only as a sensitivity. Bukalapak reports at Group level, including some overseas activity; its 2024 pair is excluded because it covers nine months of transactions against twelve of revenue. Grab's transaction measure changes basis after 2023.

Table 2. Indonesian platform evidence

This table reports the Indonesian platform evidence used in the analysis. Issuer-reported figures are disclosed by the platform; figures that require construction are derived from company-wide ratios or a third-party estimate (Table 3). The number of platform-year observations is given in parentheses, and each series covers only years for which a matched pair exists.

Platforms	Type of evidence	Years (observation)	Role
Tokopedia e-commerce	Issuer-reported, Indonesia-aligned	2022–2023 (2)	Main sample: cleanest scope, shortest window
Blibli third-party business; Bukalapak Group	Issuer-reported, broader scope	Blibli 2021–2025; Bukalapak 2020–2023 (9)	Main sample: most of the time series
Grab; Shopee Indonesia	Requires construction	Grab 2021–2023; Shopee 2022–2024 (6)	Supporting: 2023 scale comparison

4.2 Comparing Platform Activity through Time

When transaction value and revenue move sharply apart, disclosed changes in monetization, customer incentives, revenue recognition, and business scope are used to reconcile the movement where the filings permit; these reconciliations are arithmetic rather than causal.

Companies use different names for transaction activity, including gross transaction value (GTV), gross merchandise value (GMV), and total payment volume (TPV). Transaction value, V , is the umbrella concept in this proposal and is labeled *GTV*; each issuer's own label and definition is retained and checked rather than assumed to be identical.⁴

Growth rates are computed in each issuer's reporting currency, so exchange-rate movements do not enter D . US dollar values are used only for the 2023 cross-section, where rupiah figures are converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar (World Bank 2026).

4.3 Direct and Constructed Indonesia Measures

Not every company reports the same Indonesia-specific information. Table 3 sets out, for each platform, what is disclosed directly, what has to be constructed, and the limitation that follows.

⁴ Issuer labels differ. GoTo reports GTV and Sea Limited reports GMV for Shopee; for these the label differs but the measure is the same. Blibli and Bukalapak report TPV for the segments used here, and Grab reports GMV for deliveries and mobility and TPV for financial services. Where a source reports TPV the measure rests on a different basis, so it is retained under its own label rather than relabeled as GTV.

Table 3. Construction of the principal platform measures

This table sets out, for each platform, the figures disclosed directly, the figures that must be constructed, and the main limitation that follows. Indonesia GTV for Grab is estimated as Indonesia revenue $\times$ Group GTV / Group revenue, and Indonesia revenue for Shopee as Indonesia transaction value $\times$ Group monetization rate (Section 4.3). The Shopee Indonesian transaction estimate is a third-party estimate from Momentum Works (2024).

Platform	Directly available	Constructed or external input	Main limitation
Tokopedia	E-commerce transaction value and segment revenue	None for matched pair	Indonesia-aligned segment, not literal country line
Grab	Indonesia revenue; Group revenue and GTV	Indonesia GTV derived from Group figures	Indonesian ratio equals the Group ratio
Shopee	Group service monetization rate	External Indonesia transaction estimate; Indonesia revenue derived	Indonesian ratio set by the Group rate
Blibli / Bukalapak	Reported transaction and revenue pairs	None for retained pairs	Scope extends beyond Indonesian goods commerce

Grab reports Indonesian revenue directly but not Indonesian transaction value. Indonesian GTV is therefore estimated as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue} \quad (4)$$

The Group monetization rate is Group revenue divided by Group gross transaction value.

Sea Limited does not separately report Shopee's transaction value for Indonesia. Momentum Works supplies the Indonesian estimate; its annual Southeast Asian e-commerce report is widely cited for country-level platform estimates and is used because no issuer discloses the figure. The value is labeled as a third-party estimate wherever it appears, and Indonesian revenue is then estimated using Sea's Group service monetization rate:

$$\text{Indonesia revenue} = \text{Indonesia transaction value} \times \text{Group monetization rate} \quad (5)$$

Construction fixes both Indonesian ratios to company-wide relationships: Grab's E equals its Group E , while Shopee's equals the inverse of Sea's monetization rate minus one. The series contribute Indonesian levels but no independent evidence on the ratio or its movement, so they remain outside the main sample.

4.4 Evidence Beyond Platform Accounts

Table 4. Indicators used and their role

This table lists the indicators used and the role of each. The indicators answer different questions, so they are reported separately rather than pooled into one estimate. BPS denotes BPS-Statistics Indonesia. Appendix A provides variable definitions in greater detail.

Evidence source	What is observed	Role
Indonesian platform histories	Transaction value and revenue through time	Measure the wedge and its movement
BPS national e-commerce statistics	Online sales of surveyed businesses, by channel, and the number of businesses	Examine the level and sources of growth
BPS marketplace study	Business-level marketplace and recordkeeping evidence	Examine the association with financial recordkeeping
Bank Indonesia	Economy-wide digital payment measures	Context on payment growth
Platform companies outside Indonesia	Matched transaction and revenue measures	Test whether the same boundary appears elsewhere

As shown in Table 4, each indicator has a single role. The indicators answer different questions, so they are reported separately rather than pooled into one estimate.

For H2, national e-commerce value S is written as:

$$S = N \times A \quad (6)$$

where N is the estimated number of e-commerce businesses and A is implied nominal sales per business, and the change in S is divided arithmetically between the two. BPS publishes two counts of 2023 businesses; the count consistent with its stated growth figure is used, and the other is retained as a sensitivity check. The split is computed for both transitions that BPS reports, 2022–2023 and 2023–2024.

As additional analyses, BPS's published business-level comparison of marketplace and non-marketplace sellers and its split of e-commerce value by sales channel are set against the platform evidence. Bank Indonesia's economy-wide payment measures provide an independent view of digitalization (Bank Indonesia 2025). Because they cover more than e-commerce and also reflect adoption and substitution between payment methods, they are compared by growth rate and are not treated as sales estimates.

4.5 External Evidence and Robustness

The external comparison contains 48 matched annual observations for eight listed platform businesses from 2017 to 2025: eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify and Zalando. It is reported separately from the Indonesian evidence and serves as the benchmark for testing H1: a one-sided Mann–Whitney test (Mann and Whitney 1947) compares the absolute divergence $|D|$ in Indonesia with the 29 comparisons unaffected by acquisitions or changes in reporting scope, repeated with one median per platform because comparisons within a platform are not independent.

Robustness checks vary the sample-inclusion rules, revenue definitions, publication versions, and the assumptions in the constructed country measures. The resulting ranges are reported as sensitivity ranges rather than statistical confidence intervals.

5. Preliminary Evidence and Feasibility

The evidence assembled so far establishes that the proposed comparisons can be carried out and that the choice of indicator produces economically meaningful differences. The results first show the scale and movement of the wedge, then compare the platform picture with national and payment indicators.

5.1 Platform Scale and Movement

The 2023 comparison covers the three platforms introduced in Section 4.1: Shopee, Tokopedia and Grab.

Table 5. Indonesia-focused 2023 platform comparison

This table reports the Indonesia-focused 2023 platform comparison, in US\$ billions. V is transaction value, R is recognized revenue, $W = V - R$ is the invisible wedge, and $E = (V - R) / R$ is the Ecosystem Ratio (Section 3.1). Rupiah figures are converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar. Grab's transaction value is constructed from Group figures, and Shopee's transaction value and revenue are estimated; the three-platform row is the sum of the three cases, not a national total. Appendix A provides variable definitions in greater detail.

Case	Transaction value (V)	Revenue (R)	Wedge (W)	Ecosystem Ratio (E)	Evidence
Grab Indonesia	5.381	0.605	4.776	7.895x	Transaction value constructed
Tokopedia	16.331	0.405	15.926	39.296x	Issuer-reported pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Transaction value and revenue estimated
Three platforms	43.233	3.162	40.070	12.671x	Sum of three cases, not national total

Table 5 reports that cross-section. Across the three platforms, US\$43.23 billion of transaction value corresponds to US\$3.16 billion of recognized revenue, a wedge of US\$40.07 billion and a transaction-value-to-revenue ratio of 13.7 to 1 ($E = 12.7$). In terms of economic significance, the wedge is equivalent in magnitude to about 2.9 percent of Indonesia's 2023 nominal GDP of US\$1.371 trillion (World Bank 2026); the comparison is of magnitude only, since the wedge is transaction value rather than value added.

One concern is that two ratios rest on construction. Shopee's 9.0 follows from an assumed company-wide monetization rate of 10 percent, and Grab's 7.9 is its company-wide ratio, so neither provides independent Indonesian ratio evidence. To address this concern, I vary Shopee's rate from 9 to 11 percent, which puts the combined wedge between US\$39.86 billion and US\$40.29 billion, and exclude each platform in turn, which leaves a wedge of at least US\$20.70 billion.

At the 2023 exchange rate, Bilibili's and Bukalapak's reported transaction and revenue figures yield wedges of US\$3.20 billion and US\$10.50 billion. They are excluded from the total because Bilibili's measure includes online travel and Bukalapak's is reported at Group level. Their ratios, 43.4 and 36.0, sit close to Tokopedia's 39.3; the constructed ratios, by contrast, reflect company-wide monetization.

The relationship is not stable. Revenue grew faster in six of the eight main-sample comparisons, and in two comparisons revenue and transaction value moved in opposite directions. Tokopedia's 53.2 percent revenue growth against an 8.9 percent transaction-value decline gives a 62.1-point divergence. The median absolute divergence, 38.92 percentage points, compares with 7.10 points outside Indonesia, where revenue growth and transaction-value growth are closely correlated (0.98 excluding Sea, whose Shopee business also operates in Indonesia, against 0.10 in the Indonesian main sample). A one-sided Mann–Whitney test rejects equal divergence ($p = 0.002$). The result holds

after dropping the largest comparison, where revenue still grew faster in five of seven ($p = 0.006$), when Sea is excluded from the benchmark ($p < 0.001$), and with one median per platform ($p = 0.024$). Adding Grab and Shopee raises the count to ten of twelve, but their four comparisons simply reflect company-wide monetization. The evidence is therefore inconsistent with H1. Table A3 in Appendix A reports summary statistics for all available comparisons, and Figure 2 shows D for each main-sample comparison.

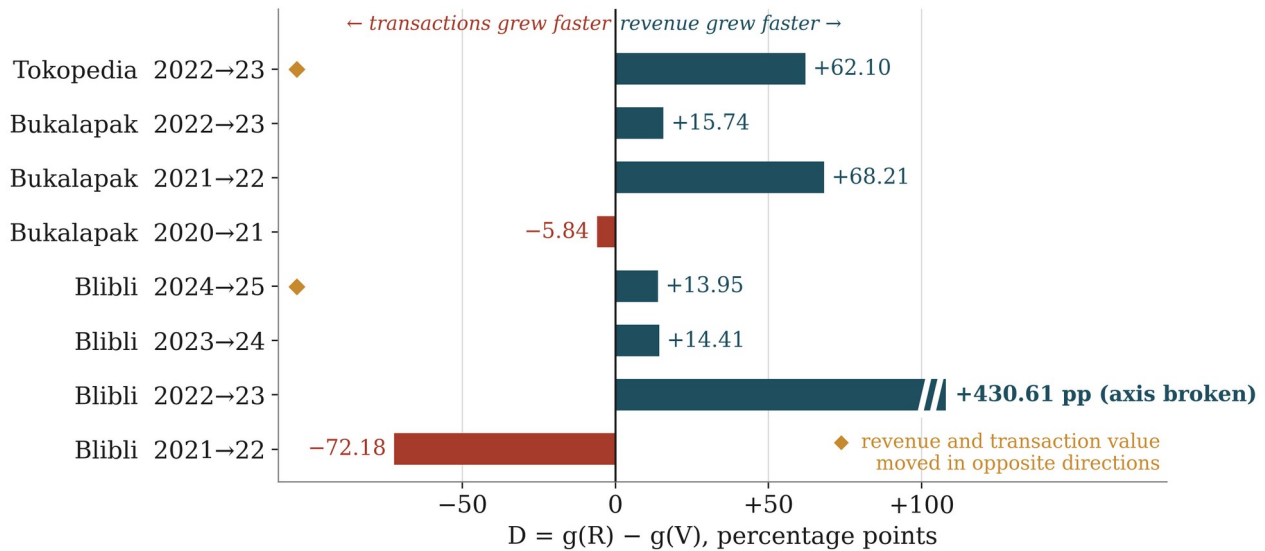


Figure 2. Growth divergence $D = g(R) - g(V)$ in the issuer-reported main sample, 2020–2025.

Positive values indicate faster revenue growth; diamonds mark the two sign reversals. Blibli 2022–2023 is shown on a broken axis.

Tokopedia illustrates the separation. In 2022–2023, transaction value fell 8.9 percent while third-party net revenue rose 53.2 percent; 60.6 percent of the revenue increase reconciles to lower customer incentives and 39.4 percent to higher gross revenue. Its accounts show growth, while its transactions show contraction. Blibli showed a similar reversal in 2024–2025: transaction value fell 1.89 percent and net revenue rose 12.07 percent. The pattern therefore appears across more than one platform and period.

5.2 Evidence Beyond Platform Revenue

National statistics give a different account of the same commerce, beginning with its level. BPS measures online sales reported by surveyed e-commerce businesses; Tokopedia reports GTV, while the external Shopee figure is a GMV estimate. In its exclusive channel split, where marketplaces account for 18.2 percent of e-commerce value, BPS places the 2023 marketplace component at Rp200.68 trillion, about US\$13.2 billion. Tokopedia's reported GTV and the external Shopee estimate together total US\$37.9 billion. Because channel attribution and reporting definitions differ, the indicators are not yet like-for-like. The study will reconcile those definitions before attributing any part of the difference to coverage.

Growth can be compared more directly. Table 6 summarizes the evidence for H2, the additional analyses and the payment context.

Table 6. Preliminary evidence beyond platform accounts

This table summarizes the preliminary evidence beyond platform accounts for H2, the additional analyses and the payment context. The evidence is drawn from BPS-Statistics Indonesia and Bank Indonesia. Growth figures refer to 2023–2024 unless another period is stated. The business-count split is arithmetic, and the recordkeeping result is an association.

Question	Preliminary evidence	Interpretation
Where does e-commerce growth come from?	E-commerce value +17.08%; estimated business count +15.31%; sales per business +1.54%. The business count accounts for 90.29% of the increase (70.95% under BPS's other count). In 2022–2023: value +40.60%, business count +27.40%, sales per business +10.36%; business-count share 70.98% (79.92%)	Growth comes mainly from more businesses selling online (H2)
Where does that growth occur?	Marketplace channel +1.45%; other channels +20.57%; 98.46% of the increase is outside the marketplace channel	Most measured growth occurs outside BPS's marketplace category (additional analysis)
Are marketplace use and financial records related?	Complete financial statements: 28.63% of marketplace users versus 12.25% of non-marketplace users	Marketplace participation is associated with stronger recordkeeping (additional analysis)
Do payment indicators show the same growth?	BPS e-commerce +17.08%; electronic-money shopping +30.47%; mobile-banking payments and purchases +82.84%; Quick Response Code Indonesian Standard (QRIS) +186.98%	Context only: payment measures grow 1.8 to 11 times as fast, and also reflect adoption and substitution

BPS also reports widespread use of non-marketplace channels. In 2023, 95.33 percent of e-commerce businesses used instant messaging. In its multiple-response measure, where a business can report several channels, social media accounted for 44.04 percent of transaction value and marketplaces for 32.74 percent; that measure is not comparable with the exclusive 18.2 percent split above. These figures describe the 2023 channel structure rather than the allocation of 2023–2024 growth. The business-count split is arithmetic and the recordkeeping result is an association. The business-count result also holds for 2022–2023, when more businesses account for about 71 percent of the increase (Table 6). Overall, the BPS evidence is consistent with H2, and the additional analyses place most national growth outside the marketplace channel, subject to the definitional differences noted above.

5.3 Feasibility and Scope of the Preliminary Findings

Overall, the preliminary evidence is inconsistent with H1 and consistent with H2: revenue did not grow in proportion to transaction value within the issuer-reported platforms, and national e-commerce growth came mainly from more participating businesses in both available transitions. The additional analyses place most of that growth outside the marketplace channel and show that marketplace sellers are more likely to keep complete financial statements.

Outside Indonesia, transaction value exceeds revenue in all eight businesses. In the latest reported year, monetization rates range from about 3 percent at Shopify to about 25 percent at Mercado Libre, and reach about 70 percent at Zalando, whose revenue includes its own retail sales. The level of the wedge therefore varies widely with the business model even where its growth is stable, which is why the test of H1 uses growth divergence rather than levels.

PMK 37/2025 provides an administrative example. It designates marketplace operators to report seller-linked transactions and withhold tax, and its implementation has been deferred. The designated operators cannot yet be mapped directly to BPS's marketplace category, and compliance and revenue effects remain outside this study.

6. Why the Problem Matters

Indonesia's digital economy is observed through platform accounts, transaction measures, national statistics and payment indicators, which disagree about scale and growth. Tokopedia's transaction value fell 8.9 percent while its revenue rose 53.2 percent, and BPS's marketplace category grew 1.45 percent while all e-commerce grew 17.08 percent. The contribution is to identify which conclusions depend on the indicator selected, and when revenue ceases to be a stable guide to the commerce a platform carries. Investors typically value platforms on revenue, while statistical agencies and tax authorities increasingly work from transaction records; when the two diverge, these users can reach opposite conclusions about the same activity.

7. Limitations and Robustness

This proposal has several limitations that also suggest avenues for future research. First, the sample is small. The main sample has three platforms, 11 platform-years and 8 comparisons, so the formal tests have limited power and rely on an external benchmark that is purposive rather than random. Second, coverage and scope are uneven: Tokopedia is observed for two years, Blibli includes online travel, and Bukalapak reports at Group level. Third, Grab's and Shopee's figures are constructed, their ratios contain no independent Indonesian information, and Shopee's revenue uses an assumed monetization rate. Finally, Section 5.2 compares indicators with different definitions; their disagreement is measured but not yet reconciled, and the national evidence covers two annual transitions for the business-count split and one for the channel split. Future research could add platforms such as Lazada and TikTok Shop once matched Indonesian revenue series become available, and could reconcile the definitions behind the BPS and platform levels.

Table 7 pairs each supported conclusion with what it does not establish.

Table 7. Main inference boundaries

This table pairs each conclusion that the evidence supports with what it does not, by itself, establish. Section 7 discusses these inference boundaries.

Evidence supports	Does not by itself establish
W , E , and D describe the size and movement of the wedge.	Participant profit, taxable income, unpaid tax, tax evasion, or missing GDP.
Issuer-reported pairs show revenue and transaction value moving apart within platforms.	Independent Indonesian ratio evidence from the constructed Grab and Shopee series.
Disclosed revenue components reconcile selected movements in the wedge.	A causal effect of incentives, monetization, or accounting choices.
BPS evidence describes the sources and channels of e-commerce growth and recordkeeping associations.	Causal business entry, productivity effects, or marketplace-induced formalization.
Platform, BPS and payment indicators disagree in level and growth.	Which indicator, if any, contains measurement error, or that payment values measure sales.
PMK 37/2025 draws on designated operators' transaction records.	Its coverage of the digital economy, or a compliance or tax-revenue effect.

One concern is that the results depend on how the sample is defined. Robustness is therefore reported as sensitivity ranges: I vary Shopee's assumed rate, exclude each platform in turn, drop the largest comparison, and restrict the longitudinal result to issuer-reported pairs.

8. Next Steps

The data and empirical framework are assembled. The remaining work will reconcile the definitions behind the BPS and platform levels, divide movements in the wedge into transaction-volume and monetization effects, extend the formal test with quarterly pairs where the filings permit, test alternative sample-inclusion rules, and complete the robustness checks. A later extension can test which indicator, or which component of revenue growth, is more informative for the stock returns of the listed platforms, subject to the small and segment-affected sample. Table 8 sets out that schedule.

Table 8. Planned work toward the final study

This table sets out the planned work toward the final study.

Period	Planned work
Proposal stage	Incorporate examination feedback; finalize the treatment of broader-scope and constructed series.
October–November 2026	Reconcile BPS and platform definitions; divide wedge movements into volume and monetization effects; complete robustness checks; finalize tables and figures.
Before the final defense	Complete the full manuscript and the final citation and consistency review.

The study treats the choice of indicator as an empirical question. It measures how far the indicators disagree, tests whether the relationship between revenue and transaction value remains stable, and identifies which conclusions about the size, growth and direction of digital activity depend on the indicator chosen.

Appendix A. Variables and Data Sources

Table A1. Variable definitions and sources

This table defines each variable used in the analysis and gives its source. V and R refer to the same period and business scope. Growth rates are computed in each issuer's reporting currency, so exchange-rate movements do not enter D . US dollar values are used only for the 2023 cross-section, converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar.

Variable	Definition	Source
V	Transaction value processed through platform i in period t : the gross value of the transactions the platform carries, labeled GTV in this proposal. Each issuer's own label, such as GMV or TPV, is retained and checked.	Issuer disclosure, or construction (Table 3)
R	Revenue that platform i recognizes in period t , for the same period and business scope as V .	Issuer disclosure, or construction (Table 3)
m	Monetization rate: the share of transaction value recognized as revenue, $m = R / V$.	Computed
W	Invisible wedge: transaction value outside the platform's recognized revenue, $W = V - R$, in currency units.	Computed
E	Ecosystem Ratio: the wedge per unit of revenue, $E = (V - R) / R$. It is constructed for this study as a descriptive transformation of the monetization rate.	Computed
D	Growth divergence: revenue growth minus transaction-value growth, $D = g(R) - g(V)$, where $g(X)$ is the percentage growth of X from one year to the next. A positive D means revenue grew faster than transaction value.	Computed from consecutive years
S	National e-commerce value: the online sales of surveyed businesses, $S = N \times A$.	BPS-Statistics Indonesia
N	Estimated number of e-commerce businesses. BPS publishes two counts for 2023; the count consistent with its stated growth figure is used, and the other is retained as a sensitivity check.	BPS-Statistics Indonesia
A	Implied nominal sales per e-commerce business, $A = S / N$. The change in S is divided arithmetically into the change in N and the change in A .	Computed from BPS-Statistics Indonesia

In Table A1, I report the definition and source of each variable used in the analysis. V and R refer to the same period and business scope, and m , W , E and D are computed from them. Table A2 traces each indicator to the filings and publications behind it, and Table A3 reports summary statistics for all available growth comparisons.

Table A2. Source lineage by evidence type

This table traces each indicator to the filings and publications behind it. Constructed series are labeled as such wherever they appear. Shopee's Indonesian transaction value is a third-party estimate (Momentum Works 2024), and Indonesian figures for Grab and Shopee are derived from Group ratios (Table 3).

Evidence type	Platform / source	Transaction-value source	Revenue / comparison source
Issuer-reported, Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Issuer-reported, broader scope	Blibli third-party business	Global Digital Niaga prospectus and results releases	Same issuer filings
Issuer-reported, broader scope	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Requires construction	Grab Indonesia	Derived from Indonesia revenue and Group monetization rate	Grab Form 20-F
Requires construction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023 and 2024; marketplace study	Business-count, channel and recordkeeping evidence
Payment statistics	Bank Indonesia	Payment System Statistics and QRIS reports	Growth-rate comparison only
Exchange rate and GDP	World Bank	World Development Indicators, 2023	Converts 2023 rupiah values; GDP scale comparison
External corroboration	eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify, Zalando	Issuer transaction measures	Issuer revenue measures
Regulatory	PMK 37/2025	Ministry of Finance / Directorate General of Taxes	Institutional architecture only

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

Table A3. Summary statistics for all available growth comparisons

This table reports summary statistics for every available year-to-year comparison. $g(V)$ and $g(R)$ are the annual percentage growth of transaction value and revenue, $D = g(R) - g(V)$, and $|D|$ is its absolute value, all in percentage points. Panel A is the issuer-reported Indonesian main sample; Panel B adds Blibli's 2020–2021 prospectus pair, which is used only as a sensitivity check; Panel C covers the 29 external comparisons unaffected by acquisitions or changes in reporting scope. Grab and Shopee are omitted because their Indonesian ratios equal company-wide ratios by construction. The external mean is dominated by Shopee's launch-stage revenue growth in 2017–2018, which is why the tests in Section 5.1 use medians and ranks. Growth rates are computed in each issuer's reporting currency. SD denotes the standard deviation.

Variable	N	Mean	Median	SD	Min	Max
<i>Panel A. Indonesian main sample (Tokopedia, Blibli, Bukalapak), 2020–2025</i>						
$g(V)$	8	30.53	17.00	46.06	-8.90	135.30
$g(R)$	8	96.41	45.74	151.38	12.07	465.33
D	8	65.87	15.07	153.58	-72.18	430.61
$ D $	8	85.38	38.92	142.18	5.84	430.61
<i>Panel B. All issuer-reported Indonesian comparisons</i>						
$g(V)$	9	28.79	14.86	43.40	-8.90	135.30
$g(R)$	9	82.57	38.28	147.55	-28.08	465.33
D	9	53.78	14.41	148.17	-72.18	430.61
$ D $	9	80.66	42.94	133.75	5.84	430.61
<i>Panel C. External benchmark: eight listed platforms, 2017–2025</i>						
$g(V)$	29	29.27	13.60	41.66	-28.43	149.93
$g(R)$	29	136.58	24.70	531.08	-15.99	2884.04
D	29	107.30	6.54	506.01	-10.10	2734.11
$ D $	29	108.69	7.10	505.70	0.36	2734.11

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